

Q.1 Scenario: Namaste Nepal Tourist App

You are building a small tourist information display for Kathmandu. The app will greet tourists and show them basic facts about Nepal — the country's name, capital city, total area in square kilometres, population, and the average temperature in Kathmandu in degrees Celsius (which can be a decimal).

Your job is to write a Java program that stores all this information in the correct variable types and prints a neatly formatted tourist fact sheet to the console.

1. Declare a variable to store the text "Nepal". Choose the correct non-primitive data type.
2. Declare a variable to store the capital city "Kathmandu".
3. Declare a variable to store the total area 147181. Use the appropriate primitive type.
4. Declare a variable to store population (30000000). Think about whether `int` can hold this number safely — use `long` if needed.
5. Declare a variable to store average temperature of nepal (18.5) Choose the correct primitive type.
6. Print all variables in a readable format using `System.out.println()`

Q.2. Everest Trekking Permit System

A trekking agency in Kathmandu is building a simple permit system for hikers heading to the Everest Base Camp trail. You have been hired as the junior developer. The manager gives you a list of details to store for each trekker and wants you to print a summary to the screen.

Trekker details to store:

Name: "Aarav Sharma" · Age: 28 · Height (cm): 172.5 · Is permit paid?: *true* · Permit ID (single letter code): 'E' · Total fee (NPR): 35000

1. Declare and assign a variable for the trekker's name using the correct data type.
2. Declare and assign variables for age, height, permit paid status, permit code, and total fee — using the most appropriate primitive type for each.
3. Print a formatted summary to the console, one detail per line with a label (e.g. **Name: Aarav Sharma**).
4. The manager also wants to know the fee in Malaysian Ringgit (1 NPR $\approx$ 0.033 MYR). Store this as a new **double** variable and print it too. Identify what type of conversion this is (implicit or explicit) and add a comment explaining why.

Q.3. ABSOLUTE CINEMA

A cinema in Pokhara sells tickets at NPR **350.75** each. At the end of the day, the manager counts **47** tickets sold. He needs two things printed:

1. The exact total revenue (with decimals).
2. The total rounded *down* to a whole number (for the cash register report).

Tasks:

- Declare an `int` for tickets sold and a `double` for the ticket price.
- Calculate the total revenue. What type does Java give the result?
- Cast the total to an `int` to get the whole-number version. Is this widening or narrowing? Why?
- Print both values with a label.

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